Table 1. PALAEOCENE FORAMINIFERAL SPECIES FROM THE LOWER SECTION (60–332 cm) OF CORE *AS* 1–9

Family	Species
Astrorhizidae	*Bathysiphon nodosariaformis* Subbotina
	Rhabdammina cylindrica Glaessner
Ammodiscidae	*Ammodiscus glabratus* Cushman and Jarvis
	Glomospira iranensis Kavary
Texturalidae	*Siphotextularia* sp.
Trochamminidae	*Trochammina quadriloba* Högland
Nodosariidae	*Dentalina nana* Reuss
	Dentalina pseudoobliquestriata (Plummer)
	Dentalina tenuissima Franke
	Lagena sulcata semiinterrupta Berry
Polymorphinidae	*Pyrulinoides acuminatus* (d'Orbigny)
Glandulinidae	*Glandulina* sp.
	Seabrookia lagenoides Ten Dam
Turrilinidae	*Buliminella parvula* Brotzen
Buliminidae	*Bulimina denticulata* Cushman and Parker
	Bulimina sp.
Discorbidae	*Pseudoparella madrugaensis* Cushman and Bermúdez
Eponididae	*Eponides plummerae* Cushman
	Eponides toulmini Brotzen
	Eponides spp.
Cibicididae	*Cibicides simplex* Brotzen
	Cibicides umbilicata Brotzen
Cassidulinidae	*Globocassidulina* sp.
Nonionidae	*Allomorphina halli* Jennings
	Chilostomella trinitatensis Cushman and Todd
	Nonionella sp.
	Pullenia jarvisi Cushman
	Pullenia quinqueloba angusta Cushman and Todd
	Quadrimorphina allomorphinoides (Reuss)
Osangulariidae	*Osangularia convexa* (Olsson)
	Pulsiphonina prima (Plummer)
	Gyroidinoides octocameratus (Cushman and Hanna)
Pleurostomellidae	*Ellipsonodosaria* cf. *paleocenica* Cushman and Todd
	Pleurostomella spp.
Anomalinidae	*Anomalinoides longi* (McLean)
	Anomalinoides midwayensis (Plummer)
	Anomalinoides umboniferus (Schwager)
	Cibicidoides proprius Brotzen
Ceratobuliminidae	*Ceratobulimina perplexa* Plummer
	Ceratobulimina tuberculata Brotzen
Globorotaliidae	*Globorotalia imitata* Subbotina
	Globorotalia varianta (Subbotina)
Globigerinidae	*Globigerina triangularis* White
	Globigerina triloculinoides Plummer

The affinities of the Palaeocene fauna in *AS* 1–9 seem to be with the circum-Atlantic and Tethys region such as the US Atlantic Coast[7], the Gulf Coast and Caribbean region[8], northern Sinai[9], Belgium[10] and the northern Alps[11]. There are particular affinities with the Swedish Palaeocene[12]. The present fauna, however, is distinct in that it has a larger variety of species of the family Pleurostomellidae and Nodosariidae and lacks species of *Nonion*, *Elphidiella*, and *Astegerina*. Differences in faunal composition between these two areas may be due to a shallower water environment for the Swedish Palaeocene, as modern representatives of species related to this Palaeocene assemblage seem to support a shelf depth for the Swedish Palaeocene and an upper bathyal depth for *AS* 1–9. The upper bathyal depth establishes the presence of a seaway in the Norwegian basin during the Palaeocene.

Sediment samples studied from different levels in the Palaeocene section of the core seem to contain essentially the same fauna. It is noteworthy, however, that in the normally magnetized Palaeocene section of the core (100–210 cm), species of *Globorotalia* are absent. As both *G. varianta* and *G. imitata* range higher up in the Palaeocene section elsewhere, the absence of *Globorotalia* in this upper section of the core may be attributed to some changes in the local environment.

In lower latitudes the Landenian is the time when an explosive evolution of marine planktonic biota took place following the drastic decline of marine planktonic life at the end of the Cretaceous[13]. The upper half of the Palaeocene section in lower latitudes is generally characterized by diversified species of *Globorotalia*, many of which possess a distinct peripheral keel. The impoverished planktonic foraminiferal population in the Norwegian Sea near 66° 30′ N. and the lack of any keeled *Globorotalia* suggest that climatic zones existed in middle Palaeocene seas.

To our knowledge this is the most northerly Palaeocene deep sea sediment recovered from the North Atlantic. Further coring in this area is indicated and should help resolve such problems as faunal diversity in high latitudes, palaeo-oceanography and the various geological and geophysical problems related to ocean floor spreading and continental drift.

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Palaeomagnetic Result from the Hook Intrusives of Zambia

THE Hook batholith is a composite granite mass intruding meta-sediments of the Katanga system in Zambia and situated at about 15° S., 26·5° E. The chief part of the batholith is syntectonic, but associated with it are numerous post-tectonic satellite bodies of diorite, syenite and gabbro[1]. An age of $500 \pm 17 \times 10^6$ yr has been reported by Snelling *et al.*[2] using the Rb–Sr method on a specimen of red granite which is also believed to be post-tectonic.

In September 1964, oriented cores were drilled at six sites in the satellite bodies: four in gabbro and one each in syenite and diorite. The natural remanent magnetization of specimens cut from the cores was measured with a spinner magnetometer[3], and alternating field demagnetization techniques[4] were employed in an attempt to isolate a primary component of magnetization. In only three of the six sites did the initially scattered directions respond to treatment, showing a grouping of directions after treatment in fields between 10 and 20 mwb.m^{-2}: higher fields led to the onset of instability and the directions became more scattered again. The mean directions obtained using the closest grouping at each site are themselves fairly well grouped, and are believed to represent the direction of thermoremanence acquired when the rocks cooled. The site means are shown in Table 1.

Palaeomagnetic poles for the three sites were combined yielding a mean North Pole position as follows: latitude 14° N., longitude 23·5° W., $K = 13$, $A_{95} = 36$, where K is the precision parameter, and A_{95} the semi-angle of the cone of confidence.

It is unlikely that results from three sites are sufficient to average secular variation, and the result should therefore be treated with some caution. But the mean pole is close to that obtained from a much more detailed study of syenitic rocks in Malawi which are of a similar age[5]. The Malawi results give those reported here greater

Table 1

Site	Latitude	Longitude	Rock type	*D*	*I*	*N*	*k*
ZA	15° 18′ S.	27° 08′ E.	Gabbro	274	+58	4	2·6
ZE	14° 49′ S.	26° 56′ E.	Diorite	320	+33	2	43
ZF	15° 09′ S.	27° 39′ E.	Gabbro	295	+55	4	6·9

D, Declination; *I*, inclination; *N*, No. of samples; *k*, precision parameter.

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significance than if they stood alone, and they jointly suggest that the extension of the African polar wander path into the Palaeozoic need no longer depend on a single result from the Silurian[6].

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Current Density Measurements in Atmospheric Electricity

IN 1929, Watson[1] pointed out that very close to the surface of the Earth the atmospheric electric current, in normal fine weather field, is carried only by positive ions, so that the current density can be computed from

$$J = E_s \lambda_{+s} \qquad (1)$$

where J is the density of the current from ionosphere to Earth, E_s is the electric field at the surface, and λ_{+s} is the conductivity of the air at the surface due to positive ions. The electric field and the conductivity, however, are measured usually at a height of approximately 1 metre. The density of the current due to conductivity, $J_c(1)$, at this height is

$$J_c(1) = E(1)\{\lambda_+(1) + \lambda_-(1)\} \qquad (2)$$

where $E(1)$ is the electric field at 1 metre and $\lambda_\pm(1)$ is the conductivity due to positive and negative ions at 1 metre. The diffusion current is usually neglected and it is assumed that the current density is approximately equal to the density of the conduction current at 1 metre, so that it can be computed from equation (2). This assumption is supported by the observations of Nolan[2,3] and Hogg[4]. It will be shown here that this assumption is not correct and that the current density should be estimated from

$$J = E(1).\lambda_+(1) \qquad (3)$$

This statement can be proved by three arguments which will be discussed here shortly.

In the first place, the observations of Hogg and Nolan do not justify the use of equation (2) to compute the current density. Measurements show that E_s and $E(1)$ are nearly equal[1,2]. In quasi-static conditions the current density is independent of height, so that it can only be computed from equation (2), if $\lambda_{+s} = \lambda_+(1) + \lambda_-(1)$, that is if the conductivity due to positive ions close to the surface is approximately twice that at a height of 1 metre. Observations of Hogg and Nolan seem to prove this, but, referring to their papers, some objections can be made. Nolan performed his observations at a place which is not representative of most atmospheric electric observatories: " . . . the station is exposed at a place between two houses, with high ground on a third side. . . ." Objections can also be made to the methods used by Hogg. Observations at various heights were performed in different seasons of the year and the apparatus was changed during the measurements. Although large corrections were applied to make the observations comparable, results obtained in this way cannot be very reliable.

Moreover, the results of Hogg and Nolan are in contradiction with the observations of Watson[1] and O'Donnell[5], who observed $\lambda_{+s} \approx 1{\cdot}15\ \lambda_+(1)$. We may therefore conclude that the observations of Hogg and Nolan do not prove the validity of equation (2) for estimating the current density.

Second, from theoretical arguments it can be shown that equation (3) is a better approximation of the current density. In 1929 Watson pointed out that space charge, transported by atmospheric turbulence, contributes to the current density. He supposed that 1 metre above the surface, transport of space charge by eddies cancels the conduction current due to negative ions, so that the current density could be estimated with equation (3). As Watson did not give the cause of this space charge, his ideas were abandoned in favour of the results of Hogg and Nolan. In my opinion, however, Watson's idea was right: negative space charge, in the form of negative ions, is transported downward neutralizing almost completely the conduction current due to negative ions. The explanation is as follows. The conditions for inactivation of ions below the level of 1 metre are about the same as the conditions above this level, if there is some atmospheric turbulence. Above the height of 1 metre the number of ions created is equal to the number of ions that is inactivated, so the net production of ions below 1 metre is equal to the additional ionization close to the surface. It can be shown that the net production below 1 metre is less than 10^6 pairs m^{-2} s^{-1}. When we assume that no negative ions are emitted by the surface, then the net production below 1 metre must be equal to the net flux of negative ions that passes this level. The flux of negative ions that drift upwards under the influence of the atmospheric electric field is of the order of 10^7 ions m^{-2} s^{-1}.

The total net production is much smaller, 10^6 ions m^{-2} s^{-1}, so that negative ions must be transported downward by eddies, almost neutralizing the conduction current due to these ions. The approximations are not valid any more when, during strong temperature inversions close to the surface, the eddy diffusivity becomes very weak. Because such strong temperature inversions do not often occur, monthly and yearly mean current densities can be computed with equation (3).

Third, the best proof of the validity of equation (3) can be found in the results of the application of this formula. The density of the current from the ionosphere to the Earth can be obtained in four different manners. (1) From the columnar resistance, r, and the potential difference between the Earth and the ionosphere, V, $J = V/r$. According to recent observations the mean potential difference is 260 kV. The columnar resistance, r, ranges from $8 \times 10^{16}\ \Omega\ m^2$ in polar regions to $24 \times 10^{16}\ \Omega\ m^2$ in polluted regions. (2) From upper air observations of the electric field and the conductivity. Above the exchange layer the current density $J = E\ (\lambda_+ + \lambda_-)$. (3) From direct current measurements performed with an isolated net at surface level. (4) From the conductivity and electric field observations at a height of approximately 1 metre: $J = E(1).\lambda_+(1)$.

In order to compare the observations of various stations the Earth was divided into three regions. I, Polar regions (Antarctic, Arctic, Greenland). II, Oceanic and continental regions above which the air is little polluted (Alaska, parts of Australia). III, Continental regions above which the air is polluted (Europe, USA).

Over fifty current density measurements at various places were compared, some of them extending over many years. The mean current densities for each group are

J in pA/m²	I	II	III
(1) Resistance: $J = V/r$	3·4	2·4	1·1
(2) Upper air: $J = E\ (\lambda_+ + \lambda_-)$	3·3	1·8	1·3
(3) Surface: direct observations	3·0	—	1·2
(4) Surface: $J = E\ (1).\lambda_+(1)$	3·4	1·9	1·4

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